SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

Pepperfry Furniture E-Commerce Database Management System

1. BASIC SQL QUERIES

1.1 SELECT QUERY

1.1.1 Display all products

This query displays all the products available in the Product table, including their product ID, product name, category ID, and other details



1.1.2 Display Selected Product Details

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------
     PRICE
----------
         1
Sofa
     27000

         2
Dining Table
     18000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------
     PRICE
----------

         3
Office Chair
      7500

         4
Bed

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------
     PRICE
----------
     30000

         5
Bookshelf
      8500
```

1.1.3 Find Product With Price Less Than ₹10,000

This query displays the products whose prices are less than ₹10,000 using the WHERE clause.

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price < 10000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
         3
Office Chair
      7500

         5
Bookshelf
      8500

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------

         6
Coffee Table
      6500

         9
Study Table
```

1.1.4 Find Product With Price Greater Than ₹10,000

This query displays the products whose prices are greater than ₹10,000 using the WHERE clause

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price > 10000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
         1
Sofa
     27000

         2
Dining Table
     18000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------

         4
Bed
     30000

         7
Wardrobe

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
     22000
```

1.1.5 Products From Lowest To Highest Price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
         6
Coffee Table
      6500

         3
Office Chair
      7500

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------

         5
Bookshelf
      8500

         9
Study Table

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
      9000
```

1.1.6 Products From Highest To Lowest Price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
         4
Bed
     30000

         1
Sofa
     27000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------

         7
Wardrobe
     22000

         2
Dining Table

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
     18000
```

1.1.7 Display Unique Product Categories

```
SQL> SELECT DISTINCT C.Category_ID, C.Category_Name
  2  FROM Product P
  3  JOIN Category C
  4  ON P.Category_ID = C.Category_ID;

CATEGORY_ID CATEGORY_NAME
----------- ----------------------------------------
          1 Sofa
          2 Bed
          3 Chair
          4 Table
          5 Storage
          6 Bookshelf
          7 Dining Table
          8 Lighting

8 rows selected.
```

2.1 Search Product Based on Price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price BETWEEN 5000 AND 15000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
         3
Office Chair
      7500

         5
Bookshelf
      8500

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------

         6
Coffee Table
      6500

         8
TV Unit

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
     12000
```

2.2 Search Products Based on Category

```
SQL> SELECT P.Product_ID, P.Product_Name, P.Price, C.Category_Name
  2  FROM Product P
  3  JOIN Category C
  4  ON P.Category_ID = C.Category_ID
  5  WHERE C.Category_Name = 'Sofa';

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_NAME
---------- --------------------------------------------------
         1
Sofa
     27000 Sofa
```

3.1 Display All Customers

```
SQL> SELECT * FROM Customer;

CUSTOMER_ID CUSTOMER_NAME
----------- ------------------------------
EMAIL
------------------------------------------
PHONE
---------------
ADDRESS
------------------------------------------
          1 Arun
arun@gmail.com
9876543210
Chennai


CUSTOMER_ID CUSTOMER_NAME
----------- ------------------------------
EMAIL
------------------------------------------
PHONE
---------------
ADDRESS
------------------------------------------
          2 Priya
priya@gmail.com
9876543211
Coimbatore


CUSTOMER_ID CUSTOMER_NAME
----------- ------------------------------
EMAIL
------------------------------------------
PHONE
---------------
ADDRESS
------------------------------------------
          3 Rahul
rahul@gmail.com
9876543212
Madurai
```

3.2 Search for a Particular Customer

```
SQL> SELECT *
  2  FROM Customer
  3  WHERE Customer_ID = 1;

CUSTOMER_ID CUSTOMER_NAME
----------- ------------------------------
EMAIL
------------------------------------------
PHONE
---------------
ADDRESS
------------------------------------------
          1 Arun
arun@gmail.com
9876543210
Chennai
```

4.1 Apply AND Condition

```
SQL> SELECT *
  2  FROM Customer
  3  WHERE Customer_ID = 5
  4  AND Customer_Name = 'Karthik';

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------
EMAIL
----------------------------------------
PHONE
---------------
ADDRESS
----------------------------------------
          5 Karthik
karthik@gmail.com
9876543214
Trichy
```

4.2 Apply OR Condition

```
SQL> SELECT P.Product_ID, P.Product_Name, P.Price, C.Category_Name
  2  FROM Product P
  3  JOIN Category C
  4  ON P.Category_ID = C.Category_ID
  5  WHERE C.Category_Name = 'Sofa'
  6  OR C.Category_Name = 'Chair';

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_NAME
---------- --------------------------------------------------
         1
Sofa
     27000 Sofa

         3
Office Chair
      7500 Chair

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_NAME
---------- --------------------------------------------------

        10
Recliner Chair
     16000 Chair
```

4.3 Apply IN Operator

```
SQL> SELECT P.Product_ID, P.Product_Name, P.Price, C.Category_Name
  2  FROM Product P
  3  JOIN Category C
  4  ON P.Category_ID = C.Category_ID
  5  WHERE C.Category_Name IN ('Sofa', 'Chair', 'Bed');

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_NAME
---------- --------------------------------------------------
         1
Sofa
     27000 Sofa

         2
Dining Table
     18000 Bed

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_NAME
---------- --------------------------------------------------

         3
Office Chair
      7500 Chair

        10
Recliner Chair
```

4.4 Apply BETWEEN Operator

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price BETWEEN 5000 AND 15000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
         3
Office Chair
      7500

         5
Bookshelf
      8500

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------

         6
Coffee Table
      6500

         8
TV Unit

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
     12000
```

5.1 Product Price Report

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------
     PRICE
----------
         6
Coffee Table
      6500

         3
Office Chair
      7500

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------
     PRICE
----------

         5
Bookshelf
      8500

         9
Study Table

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------
     PRICE
----------
      9000
```

5.2 Customer Order Report

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         O.Order_ID,
  4         O.Order_Date,
  5         O.Total_Amount
  6  FROM Customer C
  7  JOIN Orders O
  8  ON C.Customer_ID = O.Customer_ID
  9  ORDER BY O.Order_Date DESC;

CUSTOMER_ID CUSTOMER_NAME                                   ORDER_ID
----------- ---------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
         10 Nisha                                               10
10-SEP-26        19500

          9 Ajay                                                 9
09-SEP-26        37500

          8 Meena                                                8
08-SEP-26        21000


CUSTOMER_ID CUSTOMER_NAME                                   ORDER_ID
----------- ---------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
          7 Vijay                                                7
07-SEP-26        28000

          6 Divya                                                6
06-SEP-26        15500

          5 Karthik                                              5
05-SEP-26        45000
```

5.3 Product Order Report

```
SQL> SELECT P.Product_ID,
  2         P.Product_Name,
  3         OD.Order_ID,
  4         OD.Quantity
  5  FROM Product P
  6  JOIN Order_Details OD
  7  ON P.Product_ID = OD.Product_ID
  8  ORDER BY OD.Order_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
  ORDER_ID   QUANTITY
---------- ----------
         1
Sofa
         1          4

         2
Dining Table
         2          1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
  ORDER_ID   QUANTITY
---------- ----------

         3
Office Chair
         3          2

         4
Bed
```

5.4 Customers-wise Total Orders

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         COUNT(O.Order_ID) AS Total_Orders
  4  FROM Customer C
  5  LEFT JOIN Orders O
  6  ON C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.Customer_Name
  8  ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_ORDERS
----------- -------------------------------------------------- ------------
          1 Arun                                                          1
          2 Priya                                                         1
          3 Rahul                                                         1
          4 Sneha                                                         1
          5 Karthik                                                       1
          6 Divya                                                         1
          7 Vijay                                                         1
          8 Meena                                                         1
          9 Ajay                                                          1
         10 Nisha                                                         1

10 rows selected.
```

5.5 Customers-wise Total Purchase Amount

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         NVL(SUM(O.Total_Amount), 0) AS Total_Purchase_Amount
  4  FROM Customer C
  5  LEFT JOIN Orders O
  6  ON C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.Customer_Name
  8  ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------
          1 Arun
                25000

          2 Priya
                18500

          3 Rahul
                32000


CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------
          4 Sneha
                12500

          5 Karthik
                45000

          6 Divya
                15500
```